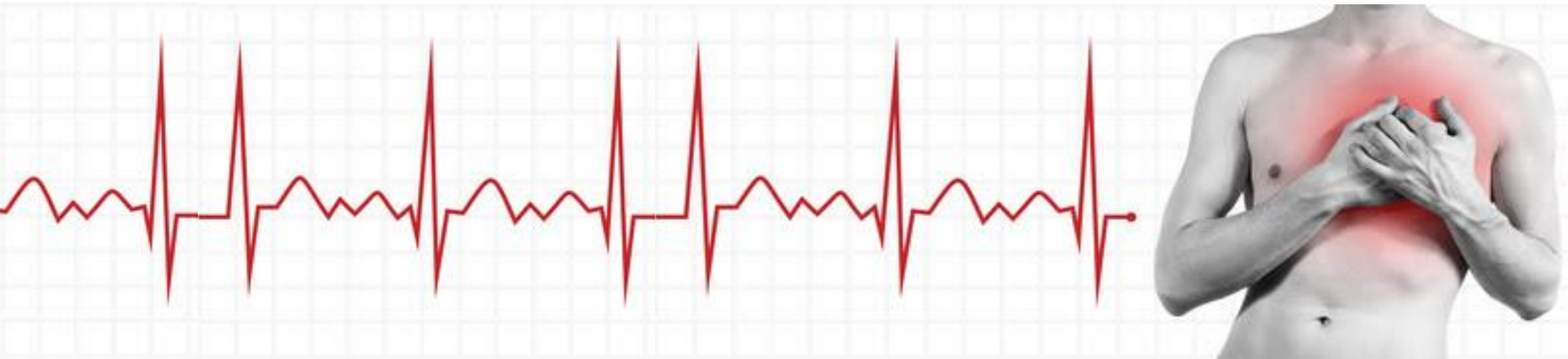


DRUG RELATED TO CARDIAC ARREST AND DYSRHYTHMIA



AKAPORN SORNKHAMPHAN
Clinical Pharmacist, Siriraj Hospital

OUTLINE

- **Introduction**
- **Drug for cardiac arrest**
- **Drug for tachycardia**
- **Drug for bradycardia**

ADVANCED CARDIAC LIFE SUPPORT (ACLS)



**2025 American Heart Association Guidelines for
Cardiopulmonary Resuscitation and
Emergency Cardiovascular Care**

Advanced Cardiac Life Support (ACLS)

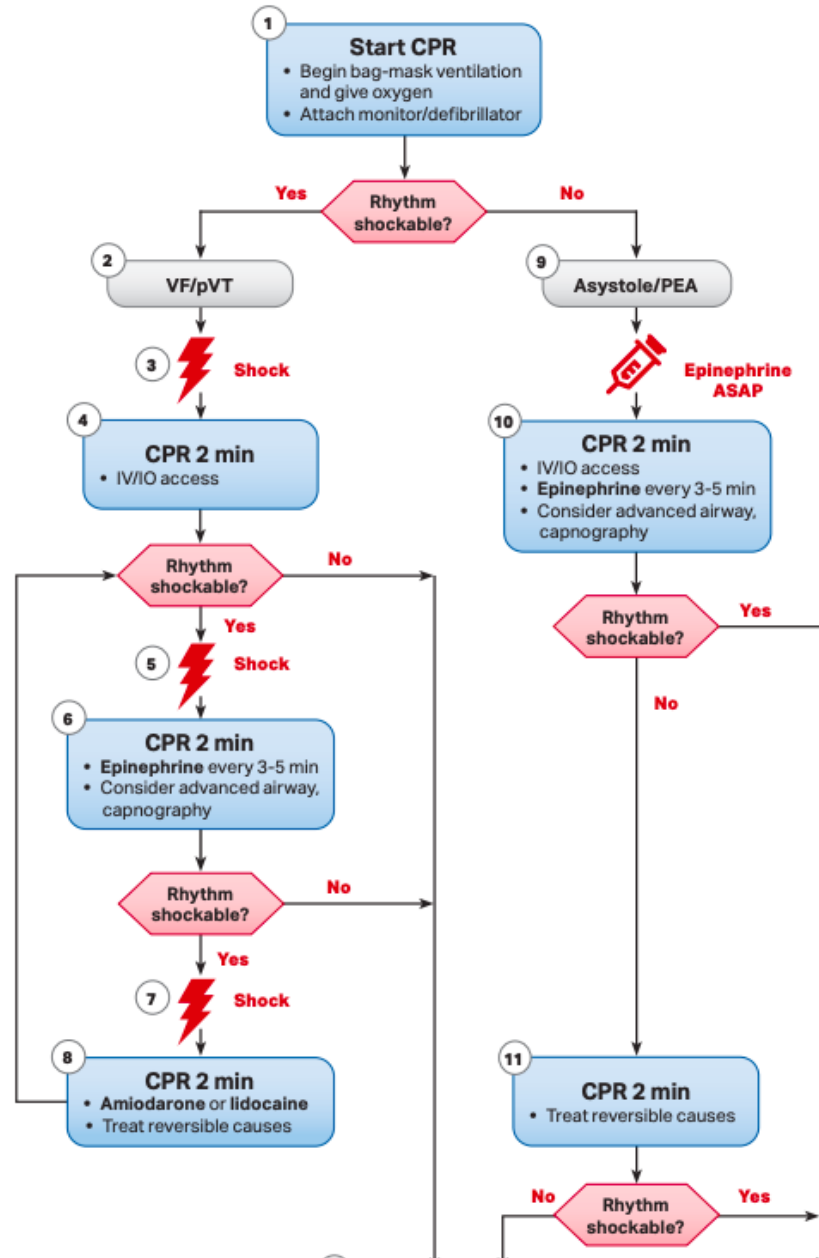
Administering adjunctive pharmacotherapy must be coordinated with nonpharmacologic treatment modality

The American Heart Association (AHA) publishes algorithms

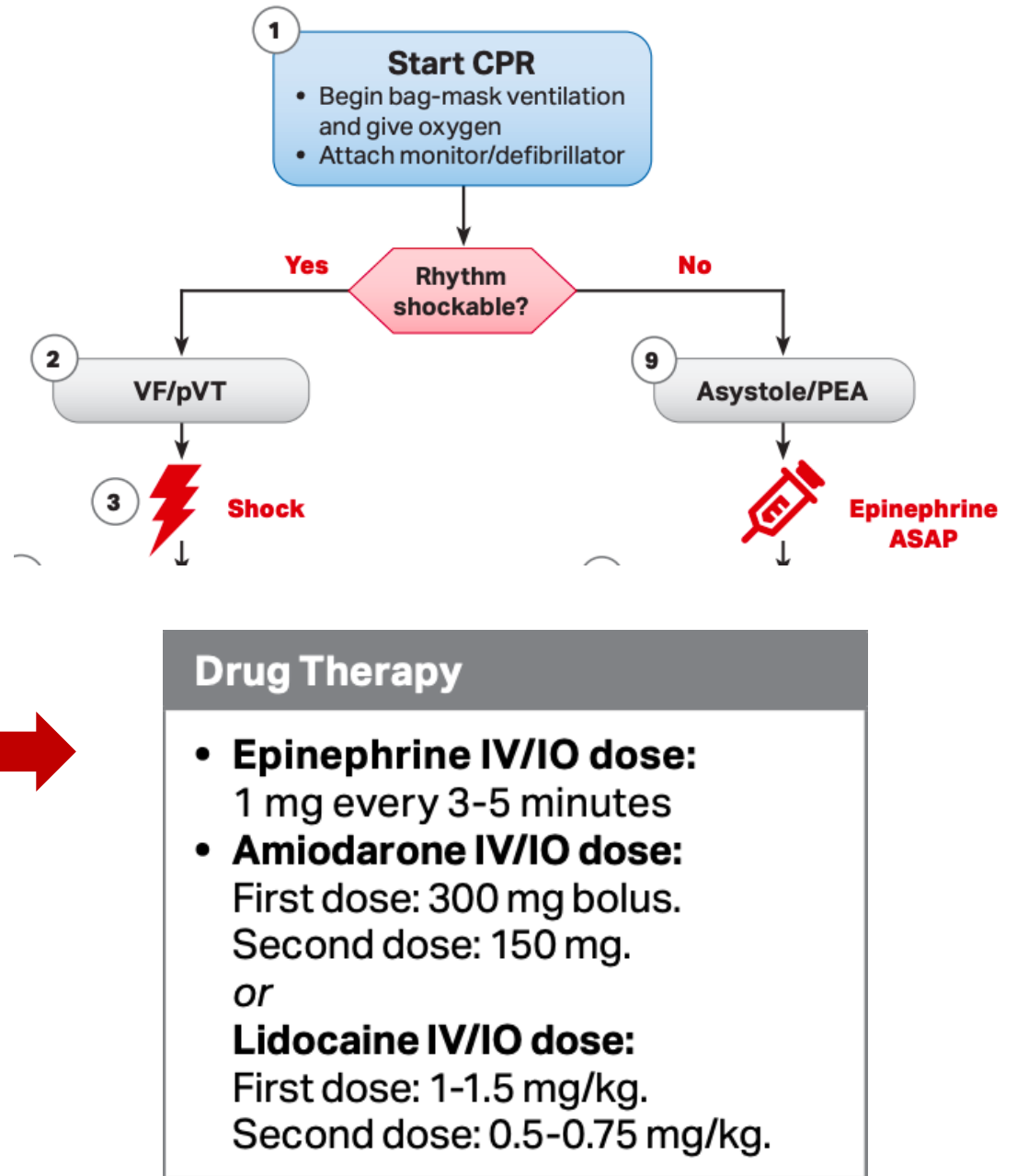
- Provide recommendations regarding the selection/timing of certain medications
- To assist with quick decision-making regarding medication administration

CARDIAC ARREST

ขั้นตอนการช่วยชีวิตภาวะหัวใจหยุดเต้นในผู้ใหญ่



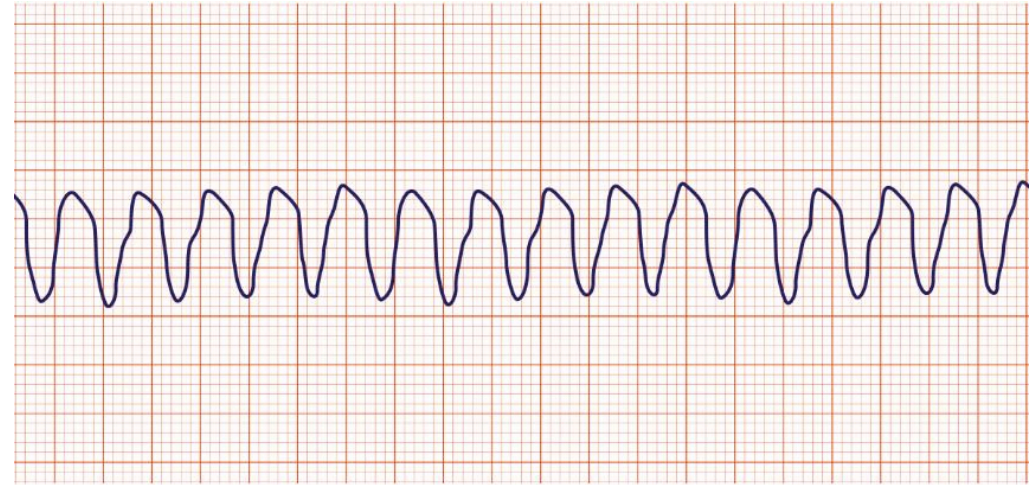
High-Quality CPR
• Push hard (at least 2 inches [5 cm]). • Push fast (100-120/min) and allow complete chest recoil. • Minimize interruptions in compressions. • Avoid excessive ventilation. • Change compressor every 2 minutes, or sooner if fatigued. • If no advanced airway, use 30:2 compression-ventilation ratio. • If advanced airway in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions. • Continuous waveform capnography – If ETCO₂ is low or decreasing, reassess CPR quality.
Shock Energy for Defibrillation
• Biphasic: Manufacturer recommendation (eg, initial dose of 120-200 J); if unknown, use maximum available. Second and subsequent doses should be equivalent, and higher doses may be considered. • Monophasic: 360 J
Drug Therapy
• Epinephrine IV/IO dose: 1 mg every 3-5 minutes • Amiodarone IV/IO dose: First dose: 300 mg bolus Second dose: 150 mg or • Lidocaine IV/IO dose: First dose: 1-1.5 mg/kg Second dose: 0.5-0.75 mg/kg
Advanced Airway
• ET intubation or supraglottic advanced airway • Continuous waveform capnography or capnometry to confirm and monitor ET tube placement
Reversible Causes
• Hypovolemia • Hypoxia • Hydrogen ion (acidosis) • Hypo-/hyperkalemia • Hypothermia • Tension pneumothorax • Tamponade, cardiac • Toxins • Thrombosis, pulmonary • Thrombosis, coronary



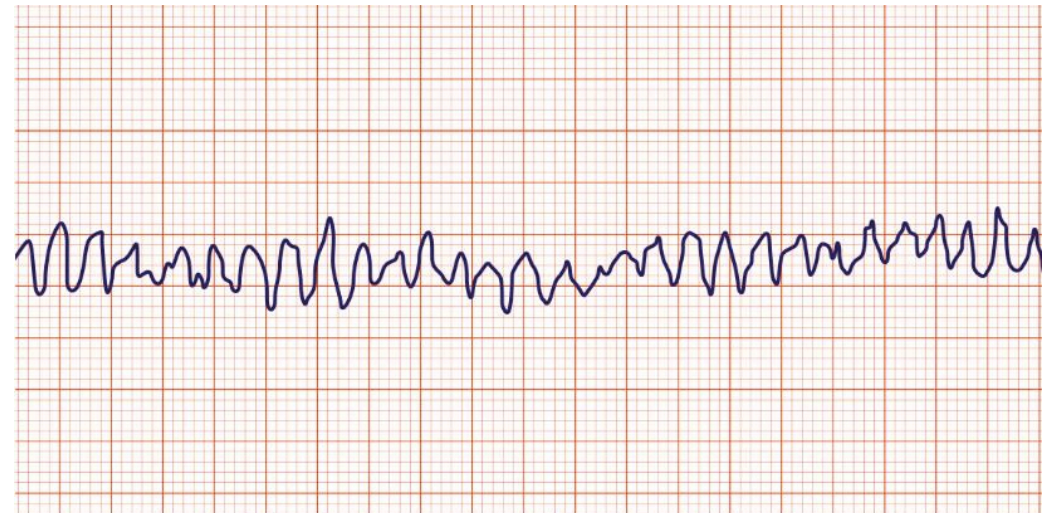
CARDIAC ARREST

Shockable rhythm

Ventricular tachycardia (VT)



Ventricular fibrillation (VF)



CARDIAC ARREST

Non-shockable rhythm

Asystole



Pulseless electrical activity (PEA)





ADRENALINE

(EPINEPHRINE)

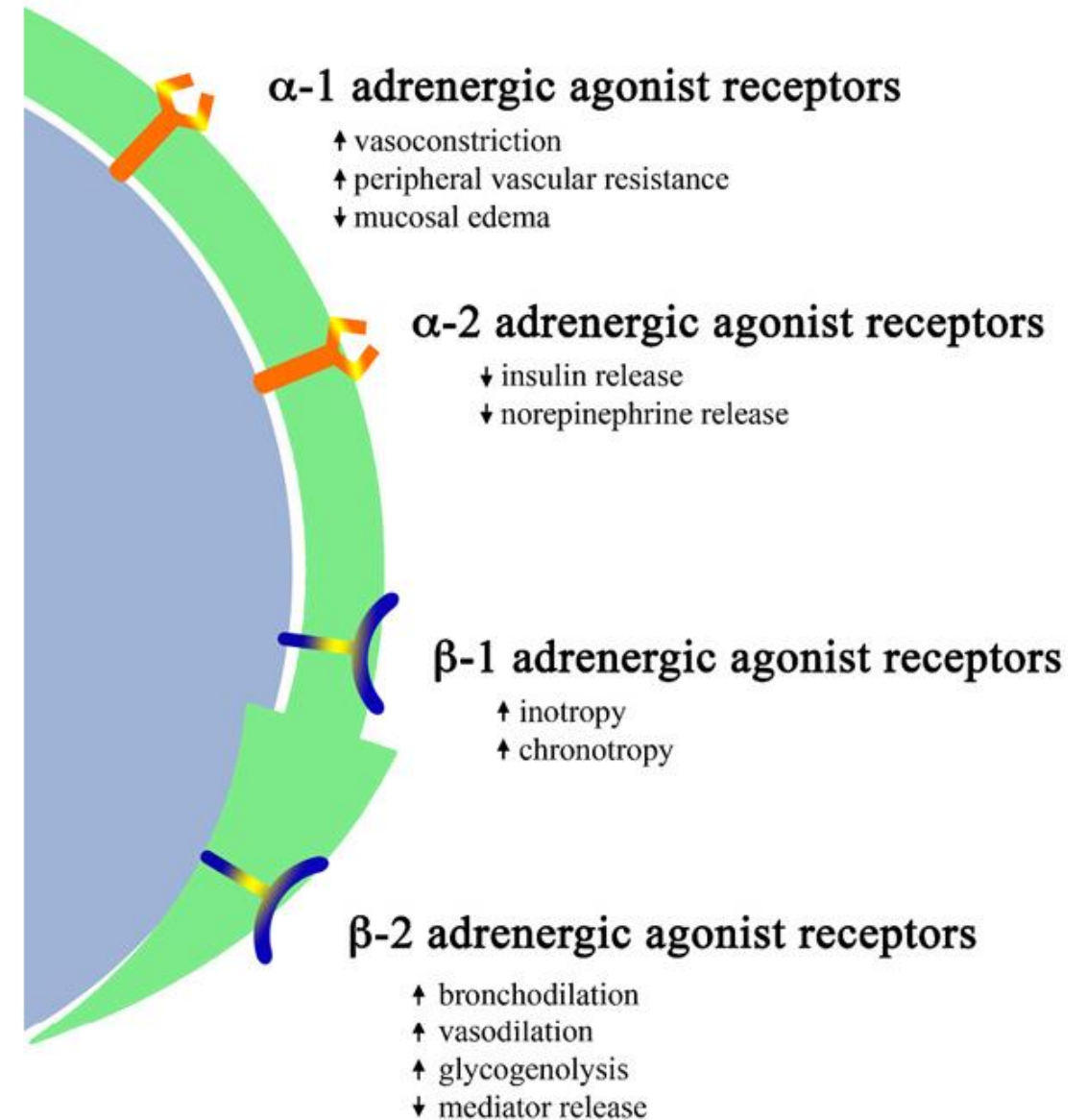
ADRENALINE

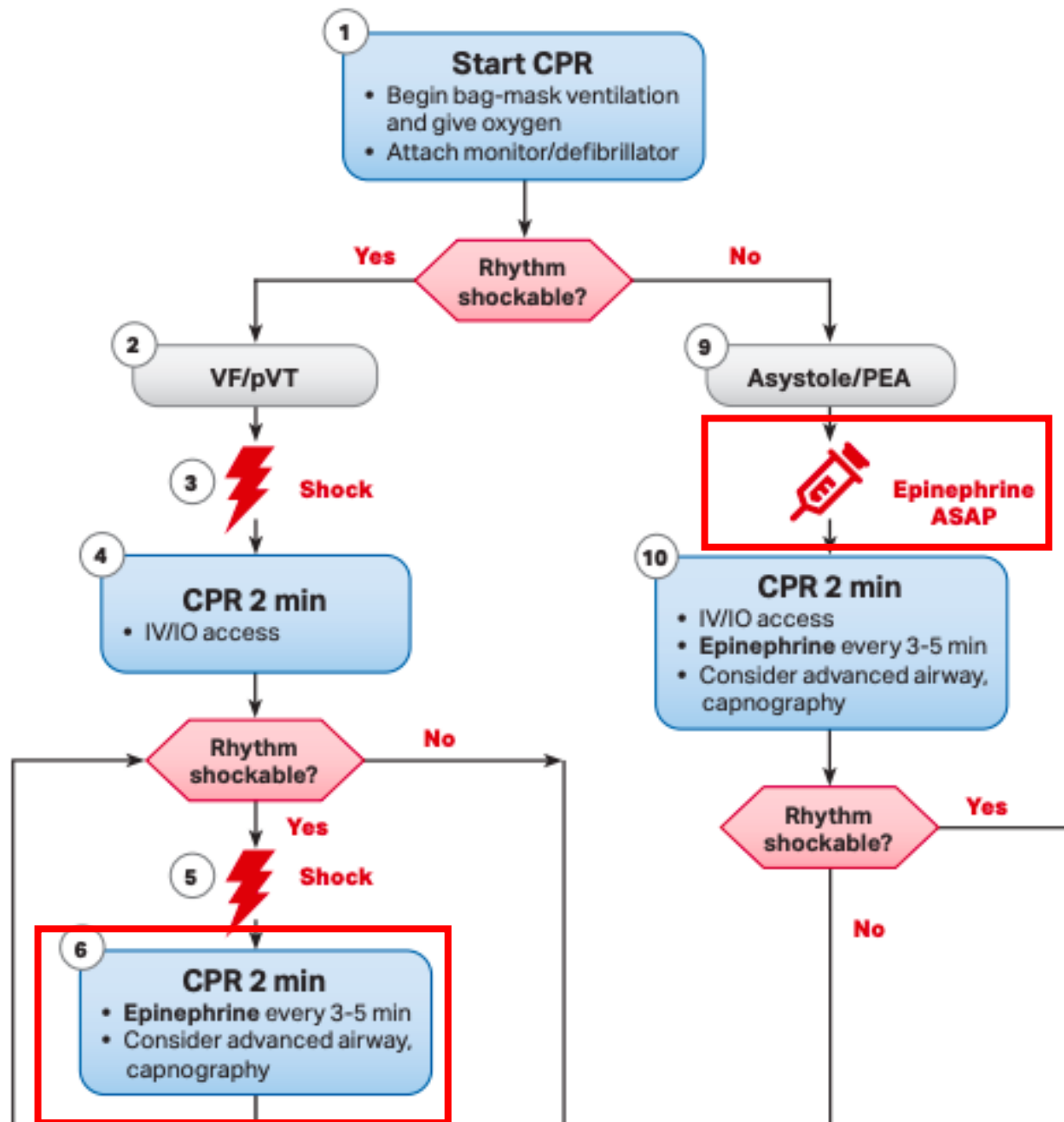
- Catecholamine
- Inoconstrictors
(vasopressors + inotropes)



เพิ่มความดันโลหิต และ อัตราการเต้นของหัวใจ

Mechanism of action





ADRENALINE

Indication

- Anaphylaxis or hypersensitivity reaction
- **Cardiac arrest**
- Hypotension or shock

Administration

Cardiac arrest

**Adrenaline 1 mg IV/IO
every 3 – 5 min**



Adrenaline
1 mg/mL

ADRENALINE

Route of Administration

Central line

- Large vein
- Rapid drug administration
- Difficult to have central line, higher skill required

Peripheral line

- Smaller vein
- 1-2 minutes needed for systemic drug distribution
- 20-mL bolus of IV fluid needed to facilitate the drug flow from the extremity into the central circulation

Route of Administration

Intraosseous (IO)

- rapid drug administration as an IV

Endotracheal tube delivery

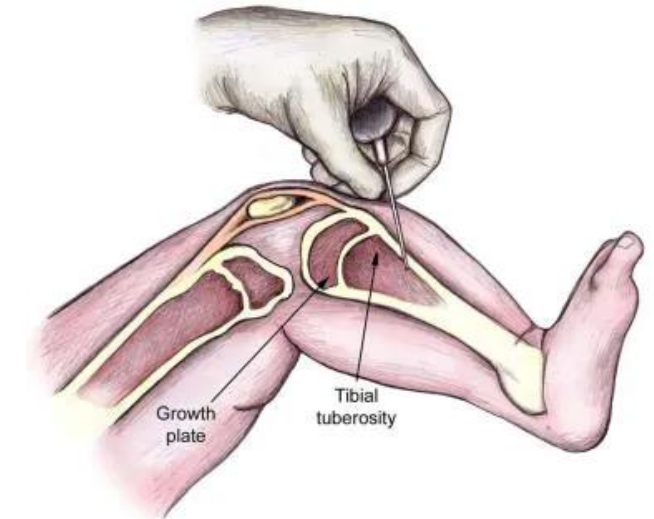
- ET dose:

Adult: 2-2.5 x of 1 mg = 2-2.5 mg

Ped: 10 x of 0.01 mg/kg = 0.1 mg/kg

- Diluents : normal saline
- Volume : Adult 10 ml, Ped 5 ml, Neonatal 2 ml
- Duration and onset: variable

ADRENALINE



Intraosseous

AMIODARONE

AMIODARONE

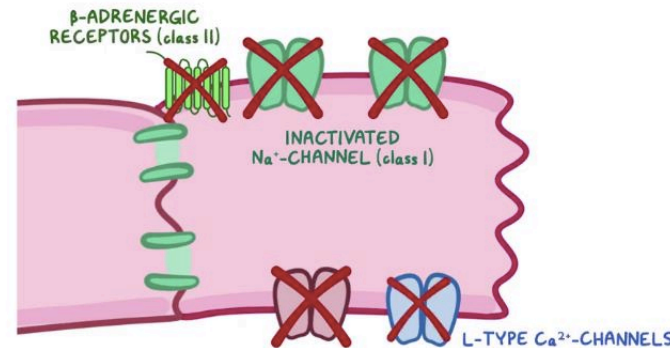
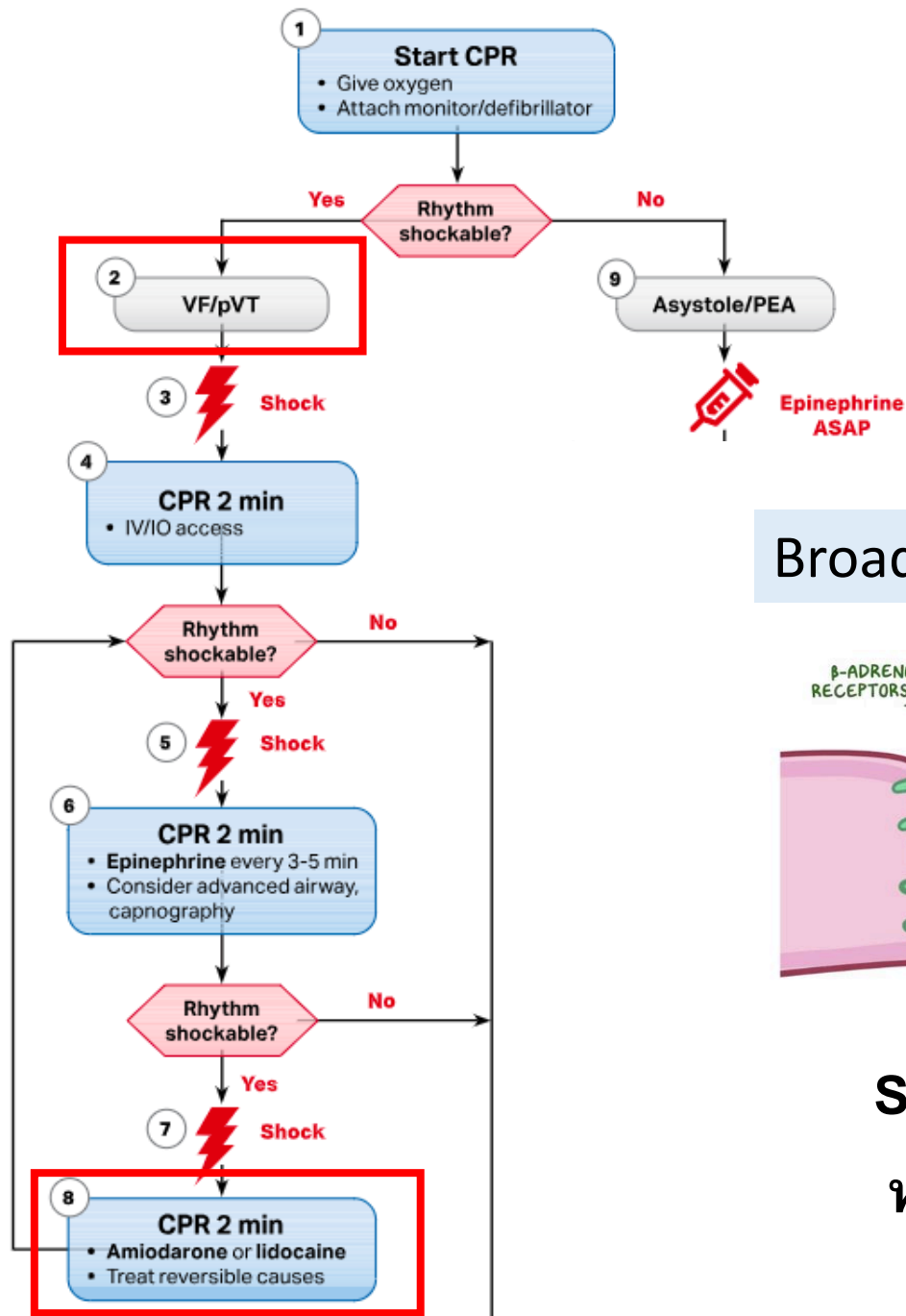
Class III : Potassium channel blocker

Broad spectrum antiarrhythmic effect & β -receptor antagonist

Indication

- Atrial Fibrillation
- Atrial Flutter
- Ventricular Tachycardia
- Ventricular Fibrillation

Do not use in 2nd or 3rd-degree heart block



Slow Heart rate
หัวใจเต้นช้าลง

AMIODARONE

Administration

Advanced cardiovascular life support : **Pulseless VT/VF**

First dose : 300 mg IV or IO **SLOWPUSH 1-2 min**
Second dose : 150 mg IV or IO every 3-5 min



Undilute then flush with NSS 20 mL
or dilute in D5W 20 mL

Amiodarone 150 mg / 3 mL

LIDOCAINE

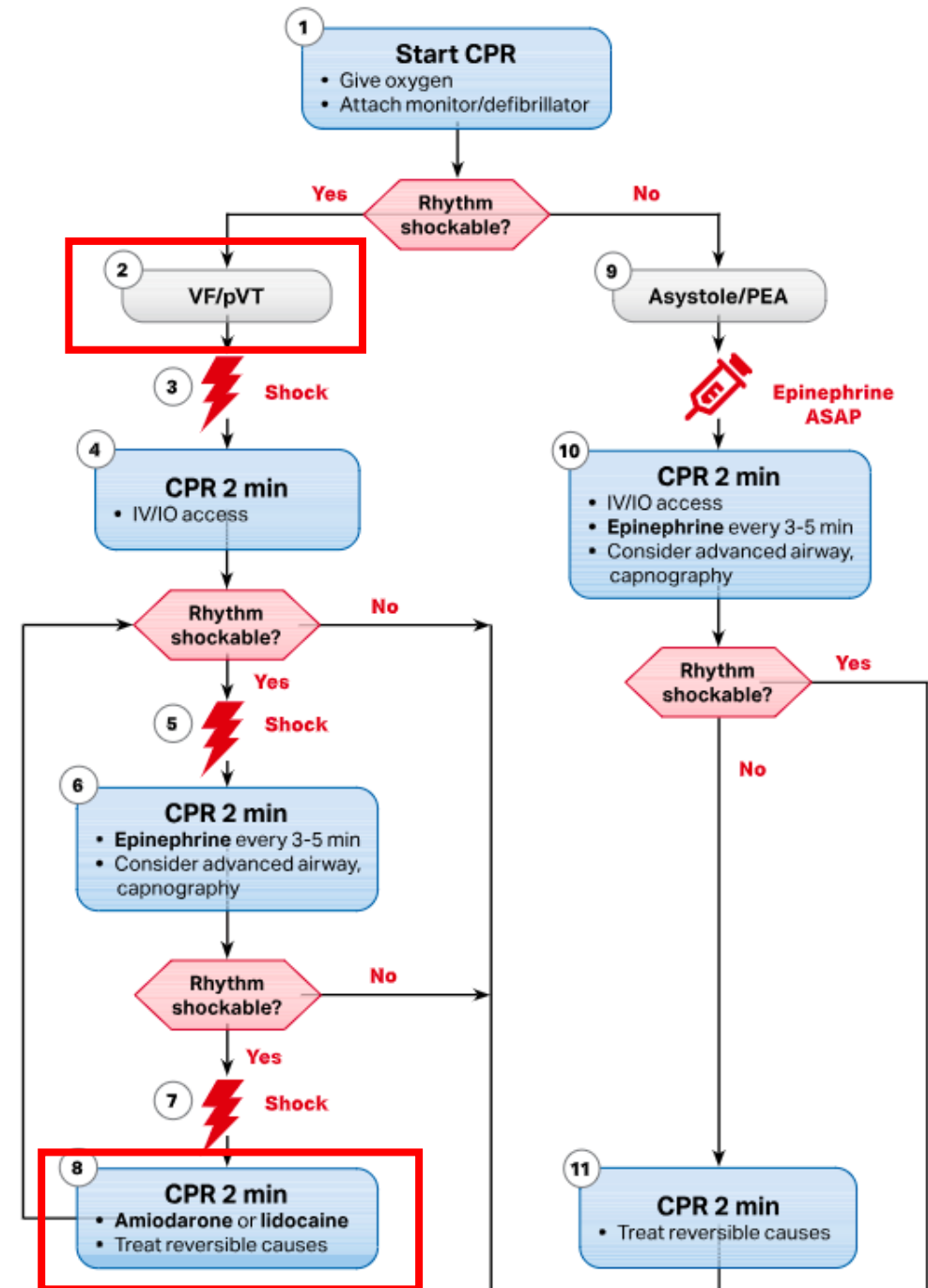
LIDOCAINE

Class Ib : Sodium channel blocker

Indication

Ventricular arrhythmias

- Ventricular Fibrillation
- Pulseless Ventricular Tachycardia (pVT)



LIDOCAINE

2% Lidocaine 2 ml

**** Preservative free ****

Administration

First dose : **1-1.5 mg/kg IV or IO**
Second dose : **0.5-0.75 mg/kg IV/IO ทุก 5-10 นาที**
(ขนาดสะสมสูงสุด 3 mg/kg ใน 1 ชั่วโมง)

Absence of IV or IO access ➡ **ET dose: 2 - 4 mg/kg**

2% Lidocaine 2 ml มีตัวยา Lidocaine 40 mg/amp

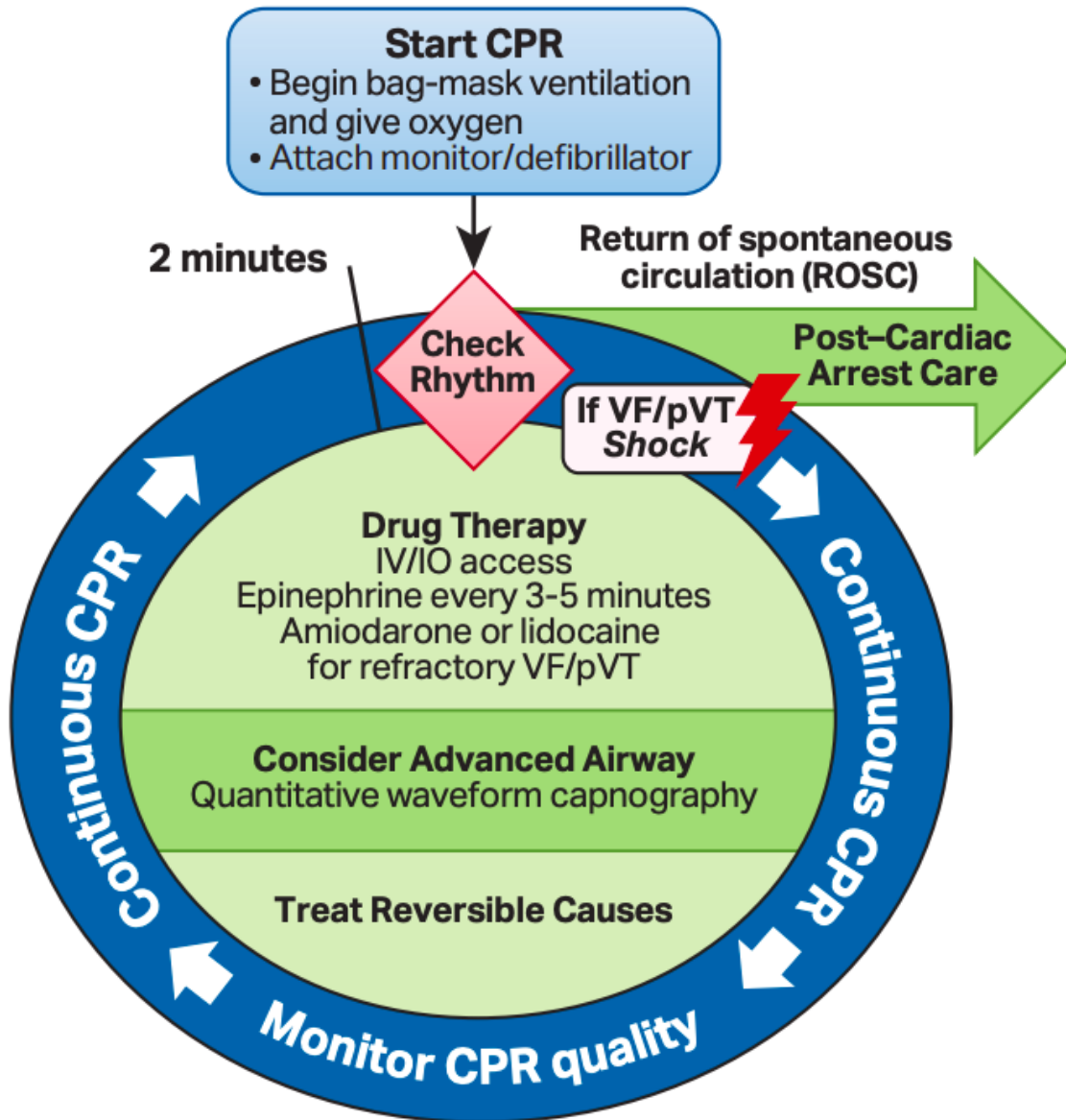


LIDOCAINE

After the occurrence of ROSC, recommended

- IV continuous infusion of **1 - 4 mg/min**
- If impaired hepatic or cardiac function or low muscle mass (such as in elderly patients): **1 - 2 mg/min**

SUMMARY MEDICATION USE



- **Epinephrine / Adrenaline (IV/IO)**

1 mg every 3 – 5 min
Optional ET : 2-2.5 mg

- **Amiodarone (IV/IO)**

First dose : 300 mg bolus
Second dose : 150 mg

- **Lidocaine (IV/IO)**

First dose : 1-1.5 mg/kg
Second dose : 0.5-0.75 mg/kg

SUMMARY MEDICATION USE

Epinephrine/adrenaline (IV/IO)

- 1 mg then flush with NSS 20 ml
- 2-2.5 mg dilute in NSS up to 10 ml via **endotracheal tube**

Amiodarone (IV/IO)

- 150-300 mg then flush with NSS 20 ml

Lidocaine (IV/IO)

- 1-1.5 mg/kg then flush with NSS 20 ml

IV Administration

- Peripheral line administration  Flush with NSS 20 ml, elevate the extremity

SODIUM BICARBONATE



CPR guideline 2020

**“Routine use of sodium bicarbonate is
not recommended for patients in cardiac arrest”**

(class III recommendation, based on level of evidence B)

Use of sodium bicarbonate in special circumstances

- **Correct metabolic acidosis**
- Hyperkalemia
- Drug overdose (Toxicity): Sodium Channel Blockers, Tricyclic Antidepressants

7.5% Sodium bicarbonate
(44.6 mEq/50mL)

SODIUM BICARBONATE



7.5% Sodium bicarbor
(44.6 mEq/50mL)

Cardiac arrest due to hyperkalemia

- Shift potassium into cells
- Dose: **50 mEq IV over 5 min**
- Repeat administration may be required, as effects are temporary

Metabolic acidosis

- Dose : **100 mEq IV over 1-2 min**
- Reassess pH, serum bicarbonate level and clinical status every 2 hours

Toxicity of drug

- **1 mEq/kg IV bolus**
- Repeat as needed until hemodynamic stability and QRS < 120 mSec

SODIUM BICARBONATE

7.5 % Sodium bicarbonate solution Osmolarity ~ 1790 mOsm/L (Hyperosmolarity)

Administration

- Hypertonic solutions >> **must be diluted** to isotonicity (1.5%) before administration
- **Undiluted** : may be given by IV injection during **cardiac arrest**
- Diluent : D5W, NSS, sterile water
- **Do not administer via ET route** >> may injury airway
- **Avoid !!** Y-site sodium bicarbonate with catecholamines (NE, Dopamine, dobutamine)
>> inactive the catecholamine

CALCIUM GLUCONATE



CPR guideline 2020

**“Routine use of calcium gluconate is
not recommended for patients in cardiac arrest”**

(class III recommendation, based on level of evidence B-NR)

**Cardiac arrest or cardiotoxicity in the presence of hypocalcemia
or hypermagnesemia**

Dose 1.5 – 3 g (15 to 30 mL of 10% solution) IV over 2 - 5 min

10% calcium gluconate 1g/10 mL

MAGNESIUM

CPR guideline 2020

“Routine use of **magnesium** is
not recommended for patients in cardiac arrest”

(class III recommendation, based on level of evidence B)

Recommendation

Polymorphic VT associated with a long QT interval
(Torsade De Pointes)

or cardiac arrest due to hypomagnesemia

Dose Magnesium sulfate 1 - 2 g IV/IO **over 15 min**
dilute in NSS 10 mL



50% MgSO_4 1 g/2mL

Circulation. 2025;152(suppl 2):S538–S577

<http://www.micromedexsolutions.com/micromedex2/librarian>

CARDIAC ARRHYTHMIAS

Bradycardia

HR < 60 bpm

- Sick sinus syndrome (SSS)
- Heart block

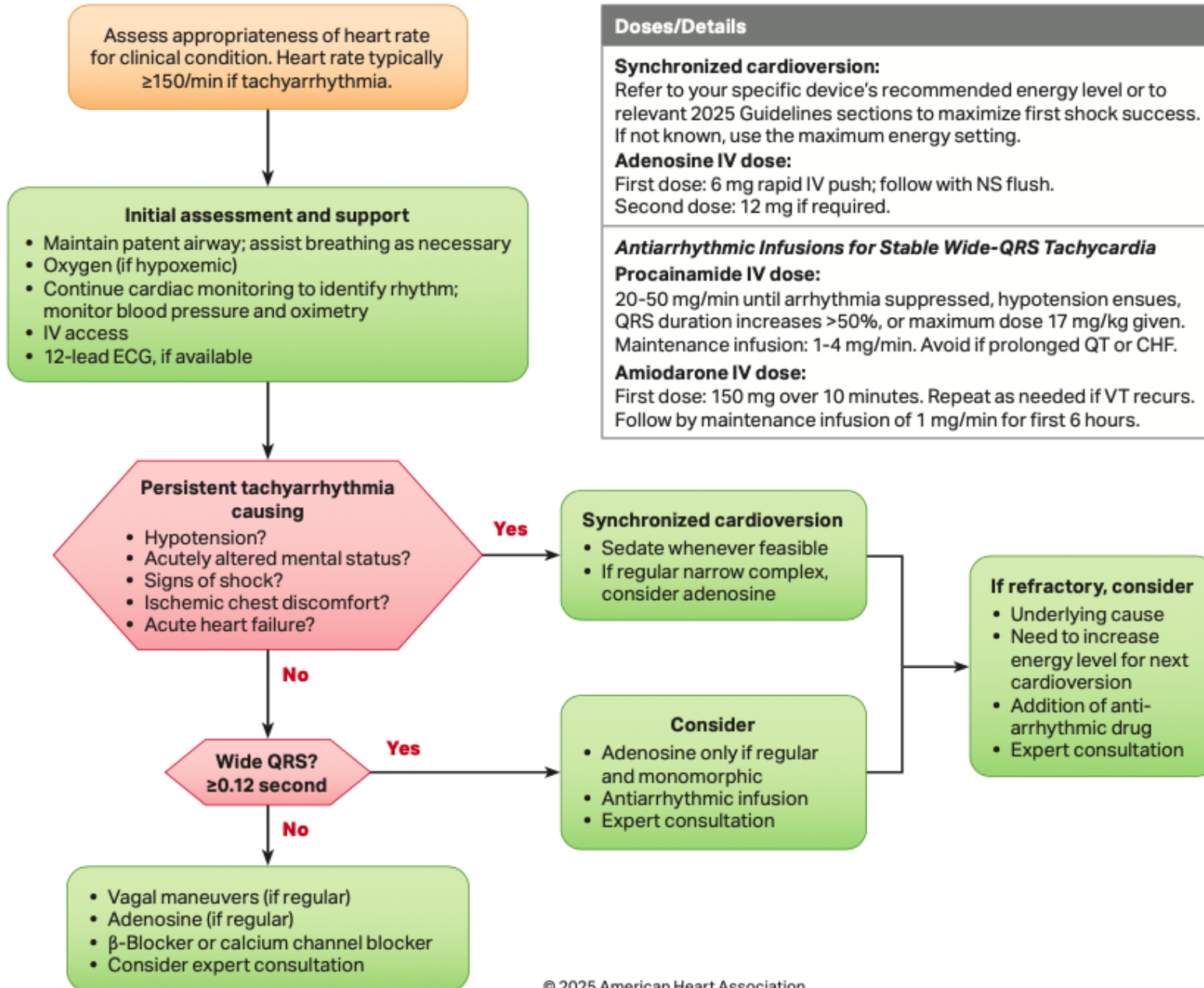
Tachycardia

HR $\geq$ 150 bpm

- Supraventricular Tachyarrhythmias
 - Sinus tachycardia, sinus arrhythmia
 - Atrial fibrillation, atrial flutter
- Ventricular Tachyarrhythmias
 - Ventricular tachycardia, ventricular fibrillation

TACHYCARDIA

Tachycardia With a Pulse Algorithm



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Tachycardia

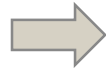
HR ≥ 150 /min



- Adenosine
- Amiodarone
- Lidocaine
- Magnesium

ADENOSINE

Recommendation



Supraventricular tachycardia (SVT)

Dose

First dose :

6 mg rapid IV push (~ 1-2 sec)

then flush NSS 20 mL (double syringe technique)

Second dose :

12 mg

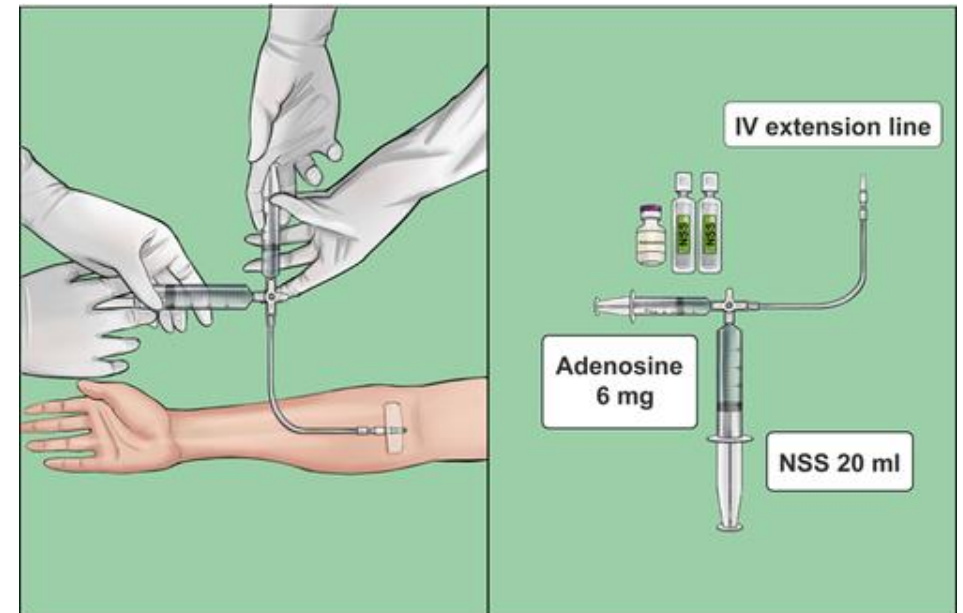
(ขนาดยาสูงสุดแต่ละครั้งไม่เกิน 12 mg)

Onset

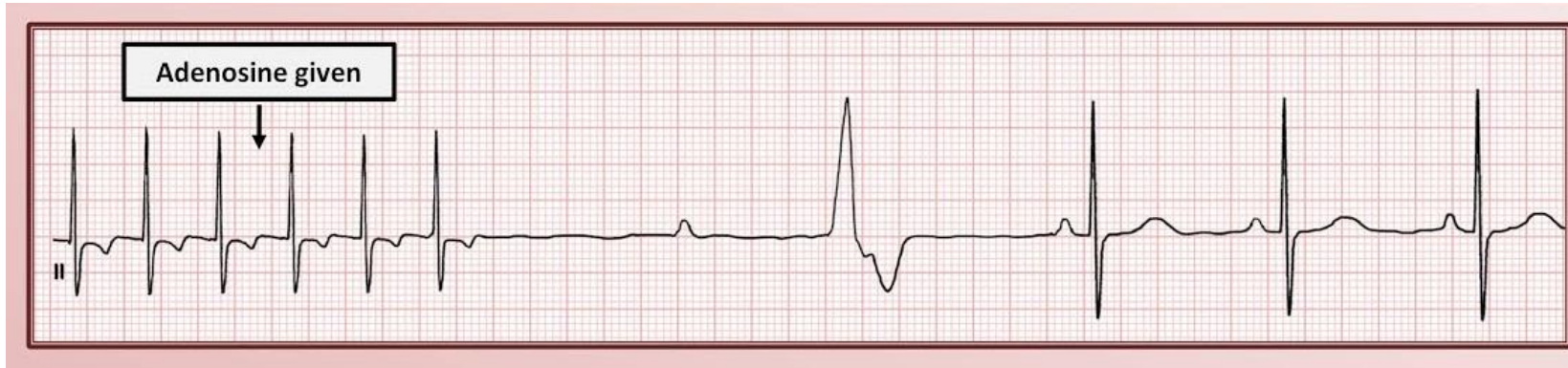
Initial response 20 - 30 seconds

Duration

< 10 seconds



ADENOSINE



Administered via **central line**

- May produce complete atrioventricular (AV) nodal blockade
- **Recommendation : Half dose 3 mg IV then 6 mg IV**

Avoid in special populations

- Asthma : transient bronchospasm
- Heart transplant recipients : may cause prolong asystole
- Wolff-Parkinson-White (WPW)



Circulation. 2025;152(suppl 2):S538–S577

<http://www.micromedexsolutions.com/micromedex2/librarian>

ADENOSINE

Adverse effects

- Flushing (18%)
- Dyspnea (12%)
- Chest discomfort (common)
- Asthmatic and bronchospasm  Use aminophylline to reverse

Storage

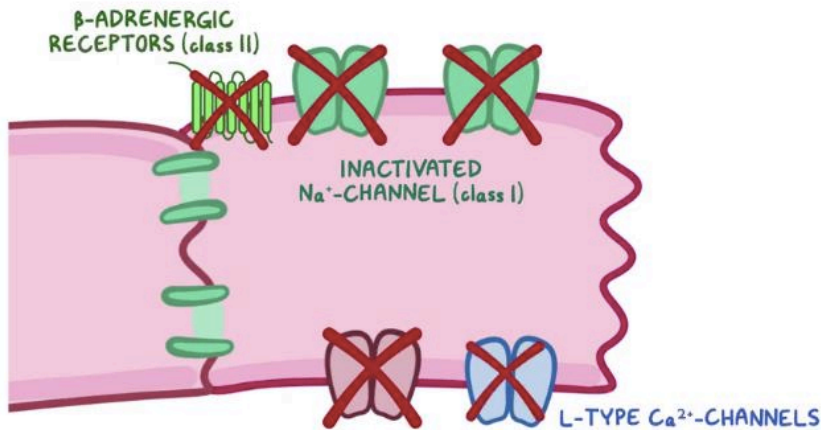
Do not refrigerate  Crystallization

** If crystallization occurs, allow solution to warm to room temperature to dissolve the crystals prior to use

AMIODARONE

Class III : Potassium channel blocker

Broad spectrum antiarrhythmic effect & β -receptor antagonist



Slow Heart rate

หัวใจเต้นช้าลง

Indication

- Atrial Fibrillation
- Atrial Flutter
- Ventricular Tachycardia
- Ventricular Fibrillation

Administration

AMIODARONE

Rate control (ควบคุมอัตราการเต้นของหัวใจ)

ขนาดเริ่มต้น 300 mg IV over 1 hr then 10-50 mg/hr IV over 24 hr

Pharmacologic cardioversion

Amiodarone 150 mg IV over 10 min then 1 mg/min (360mg) in 6 hr
then 0.5 mg/hr (540 mg) in 18 hr

Sinus rhythm maintenance in AF (IV)

Amiodarone 150 mg IV over 10 min then 1 mg/min (360mg) in 6 hr then 0.5 mg/hr
(540 mg) in 18 hr หลัง 24 hr ลดขนาดลงเป็น 0.25 mg/min

Life-threatening Ventricular arrhythmia

Amiodarone 150 mg IV over 10 min และสามารถให้ซ้ำได้ทุก 10 min
then 1 mg/min (360mg) in 6 hr then 0.5 mg/hr (540 mg) in 18 hr
Max 2.2 g/24 hr

AMIODARONE

Administration



Amiodarone
150 mg/3mL



5% DW เท่านั้น !!

Max concentration

- Peripheral : $\leq 2 \text{ mg/mL}$
- Central : $\leq 6 \text{ mg/mL}$

Stability

- 24 hour

**** กรณีใช้ IV set ที่เป็น PVC แนะนำบริหารยาไม่เกิน 12 ชั่วโมง ****

Side effects

AMIODARONE



Amiodarone Injection

Loading dose เช่น Amiodarone 150 mg in 10 min



ยับยั้ง Alpha receptor



**ความดันต่ำ
(Hypotension)**

Max rate < 15 mg/min

ยกเว้น Cardiac arrest ต้องให้ rapid bolus

Polysorbate



เกิดในยาฉีดเท่านั้น

สารละลายยา

Benzyl alcohol



**หลอดเลือดดำอักเสบ
(Phlebitis)**

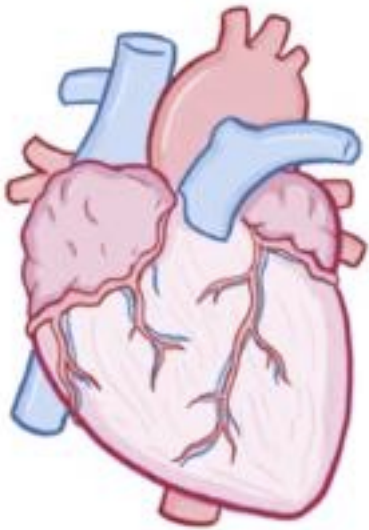
Max concentration

Peripheral : ≤ 2 mg/ml

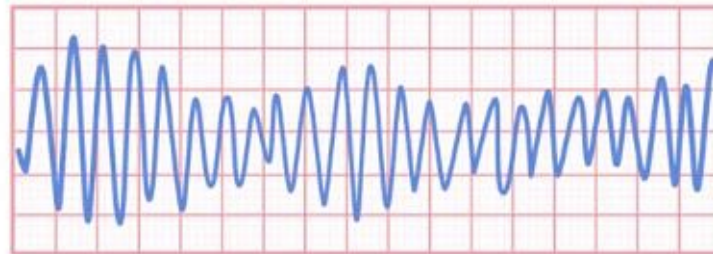
Central : ≤ 6 mg/ml

AMIODARONE

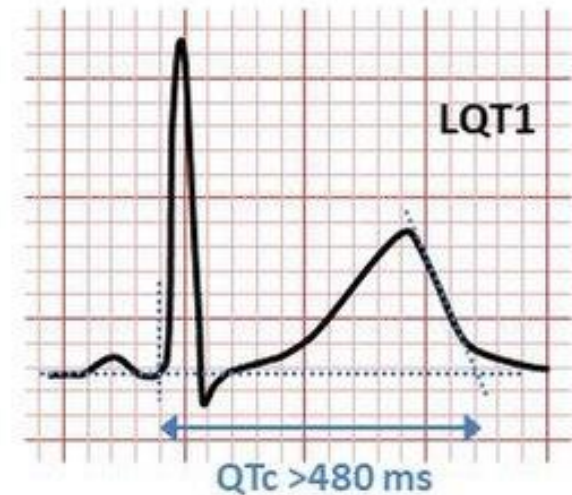
Side effects



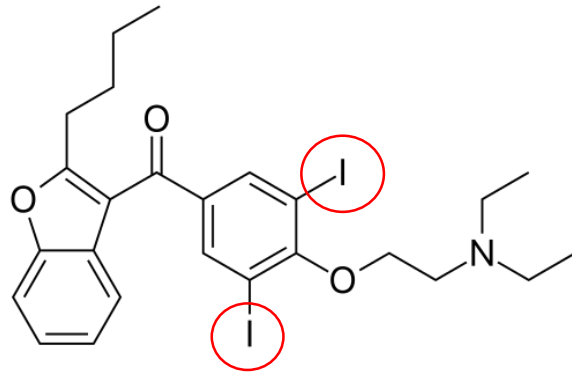
- หัวใจเต้นช้า (Bradycardia)
- ภาวะหัวใจล้มเหลว
- Torsade de pointes



- Prolong QT interval

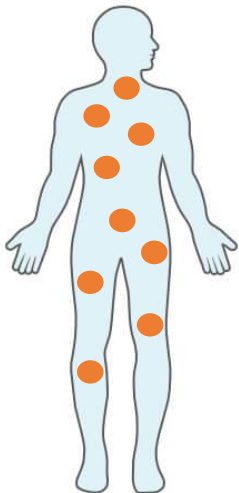


AMIODARONE



Amiodarone

ตัวยาละลายในไขมันได้ดี (Lipophilic)



Half life ~ 26 -107 วัน

Side effects

- ภาวะไทรอยด์ต่ำหรือสูงเกิน (Hypo/Hyperthyroid)
- การทำงานของตับผิดปกติ (Liver dysfunction)
- ตาสู้แสงไม่ได้
- ผิวหนังไวต่อแสง (Photosensitivity)

Dose-dependence side effects

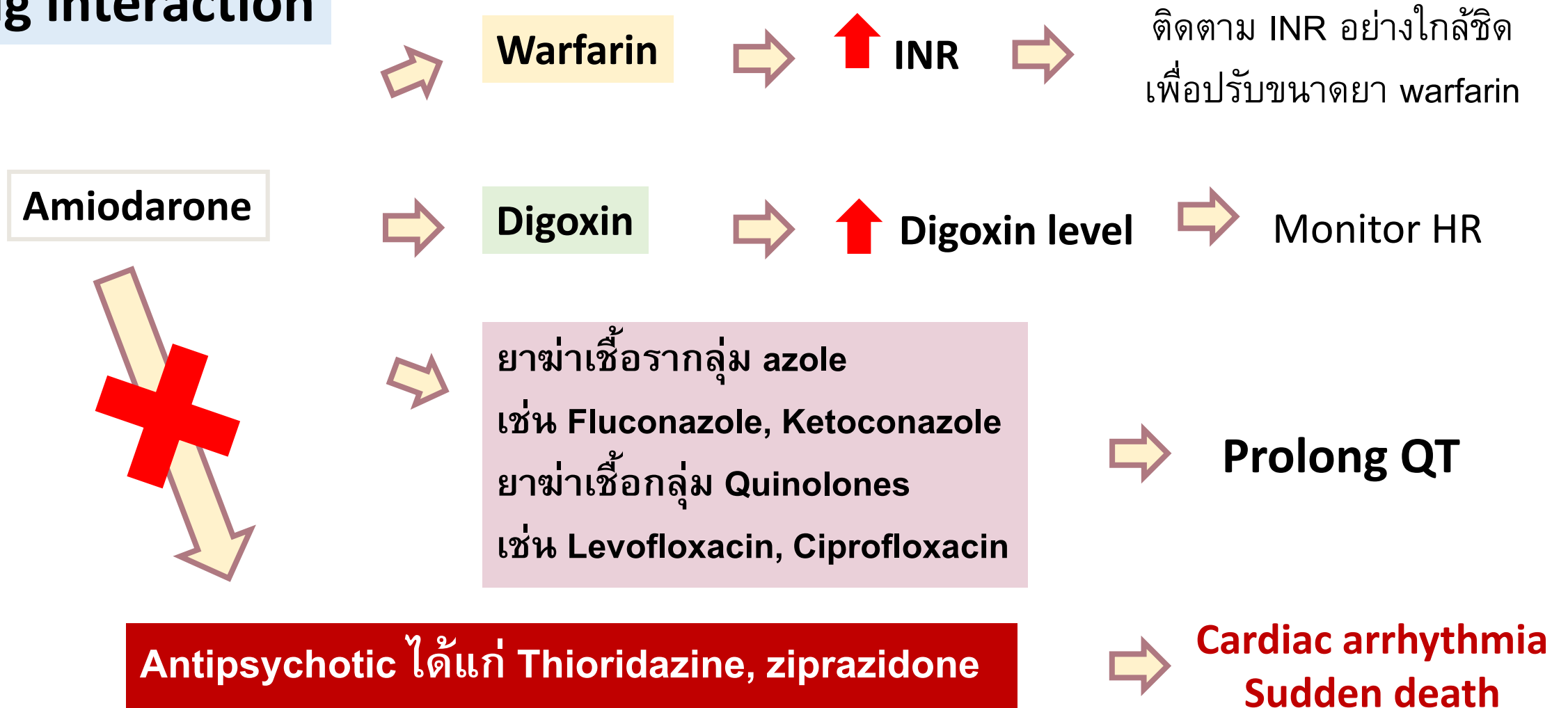
- พังผืดที่ปอด (Pulmonary fibrosis)
- การสะสมของยาในกระจกตา (Cornea deposits)
- ผิวหนังเกิดสีผิดปกติ (Blue-gray skin discoloration)

ก่อนเริ่มใช้ยาควรมีการติดตาม

Thyroid function test, Liver function test

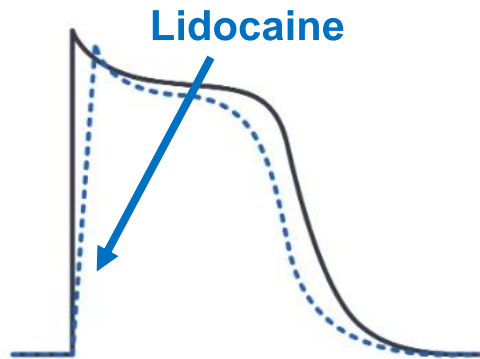
AMIODARONE

Drug interaction



LIDOCAINE

Class Ib : Sodium channel blocker



- ยับยั้ง action potential ใน phase 0
- ทำให้การส่งกระแสไฟฟ้า (conduction) ช้าลง
- ไม่มีผลยับยั้ง K^+ channel จึงไม่เกิด QTc prolongation

Indication

Ventricular arrhythmias

- Ventricular Fibrillation
- Pulseless Ventricular Tachycardia (pVT)

2% Lidocaine 2 ml
**** Preservative free ****



LIDOCAINE

Dose

Loading dose : 1-1.5 mg/kg IV
then 0.5-0.75 mg/kg IV every 3 - 5 min
(Max 3 mg/kg in 1 hour)

Maintenance dose : 1 - 4 mg/min

** Impaired hepatic or cardiac function or low muscle mass : **1 - 2 mg/min** **

Administration

Diluent : NSS, 5%DW, D5S/2, Lactate ringer
Max concentration : **8 mg/ml (8:1)** ให้ทาง peripheral line ได้
Stability : 24 hours

**** ขณะให้ Lidocaine ต้องมีการ monitor EKG ตลอดเวลา ****

การผสมยา 2% Lidocaine 2 ml สำหรับ maintenance infusion

1 amp มียา 40 mg/2 mL

ความเข้มข้นของสารละลายที่ต้องการเตรียม	Drip rate (mL/hr)			
	1 mg/min	2 mg/min	3 mg/min	4 mg/min
2 mg/mL ผสมยา 200 mg ใน 100 mL ใช้ยา 5 amp + สารน้ำจนครบ 100 mL	30	60	90	120
4 mg/mL ผสมยา 400 mg ใน 100 mL ใช้ยา 10 amp + สารน้ำจนครบ 100 mL	15	30	45	60
8 mg/mL ผสมยา 400 mg ใน 50 mL ใช้ยา 10 amp + สารน้ำจนครบ 50 mL	7.5	15	22.5	30

สารน้ำที่เข้ากันได้ : 5%DW, NSS, D5S/2, Lactate ringer

LIDOCAINE

Side effects

- ชาบริเวณรอบปาก
- พูดไม่ชัด
- คลื่นไส้ อาเจียน
- กระวนกระวาย
- ชี้น
- ชัก เกร็งกระตุก

Caution

ระวังการใช้ในผู้ป่วยภาวะตับและไตบกพร่อง

หากมีอาการผิดปกติ แจ้งแพทย์ และหยุดยาทันที



MAGNESIUM SULFATE

Polymorphic VT associated with a long QT interval (**Torsade De Pointes**)

Magnesium sulfate 1 - 2 g IV/IO **over 15 min** (dilute in NSS 10 mL)

Paroxysmal supraventricular tachycardia

(Failing to respond to simpler treatments and without evidence of myocardial damage)

Magnesium sulfate 3 – 4 g IV/IO **over 30 sec**
dilute in NSS 40 mL with extreme caution

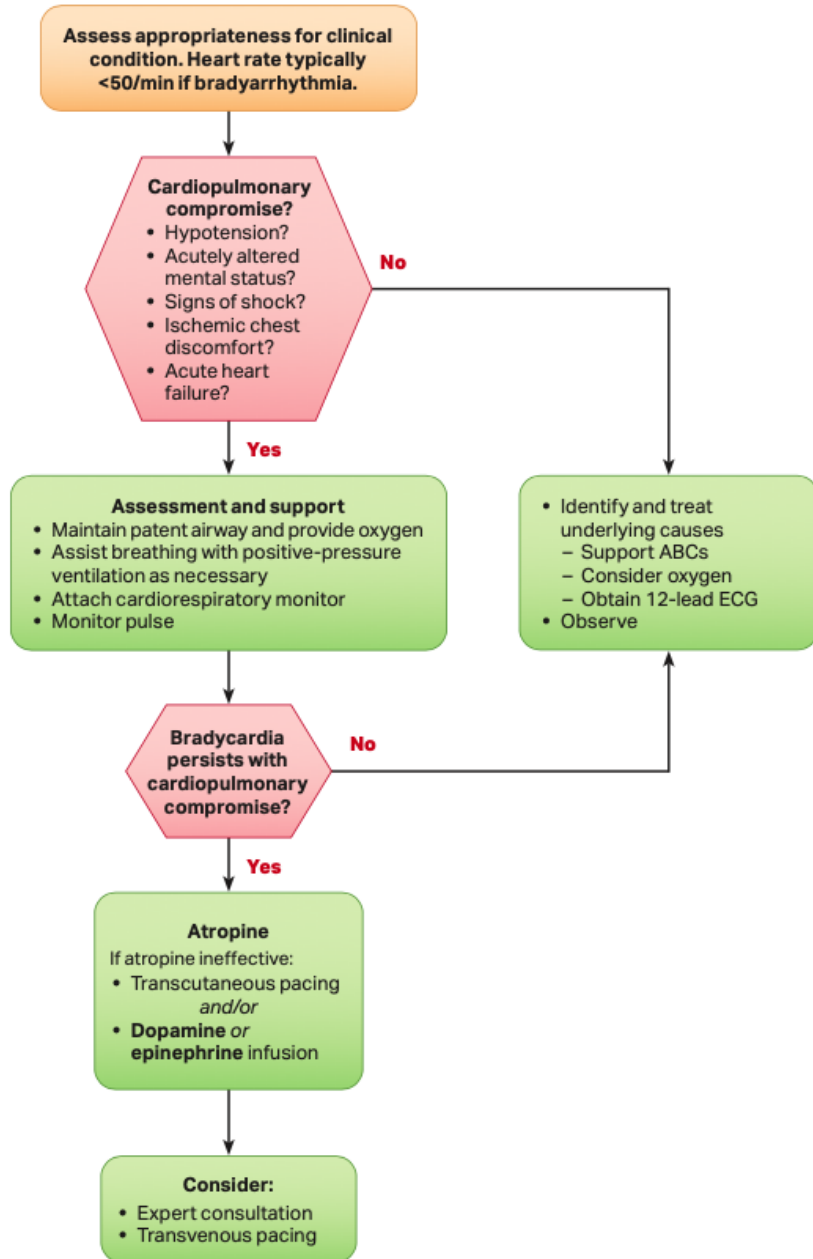
- Rate IV administration **should not exceed 2 g/h** (Hypotension)



50% MgSO₄ 1 g/2mL

BRADYCARDIA

Bradycardia With a Pulse Algorithm



Doses/Details
Atropine IV dose: First dose: 1 mg bolus. Repeat every 3-5 minutes. Maximum total dose: 3 mg.
Dopamine IV infusion: Usual infusion rate is 5-20 mcg/kg per minute. Titrate to patient response; taper slowly.
Epinephrine IV infusion: 2-10 mcg per minute infusion. Titrate to patient response.
Possible Causes
• Myocardial ischemia/infarction • Drugs/toxicologic (eg, calcium-channel blockers, β-blockers, digoxin) • Hypoxia • Electrolyte abnormality (eg, hyperkalemia)

Bradycardia

HR < 60 /min



- Atropine
- Dopamine
- Epinephrine/adrenaline

ATROPINE

Inhibits muscarinic acetylcholine receptors (Ex. smooth muscle, cardiac tissue)
leading to an **increase** in **cardiac output and heart rate**

Symptomatic bradycardia

Dose

Atropine 1 mg IV/IO bolus repeat every 3 – 5 min

Maximum total dose 3 mg (0.5 mg x 6 amp)

Thailand

Atropine 0.6 mg IV/IO bolus repeat every 3 – 5 min

Maximum total dose 3 mg (0.6 mg x **5 amp**)

ET route: Dilute 1 - 2 mg in ≤ 10 mL of sterile water or NSS



Atropine 0.6 mg/mL

ATROPINE

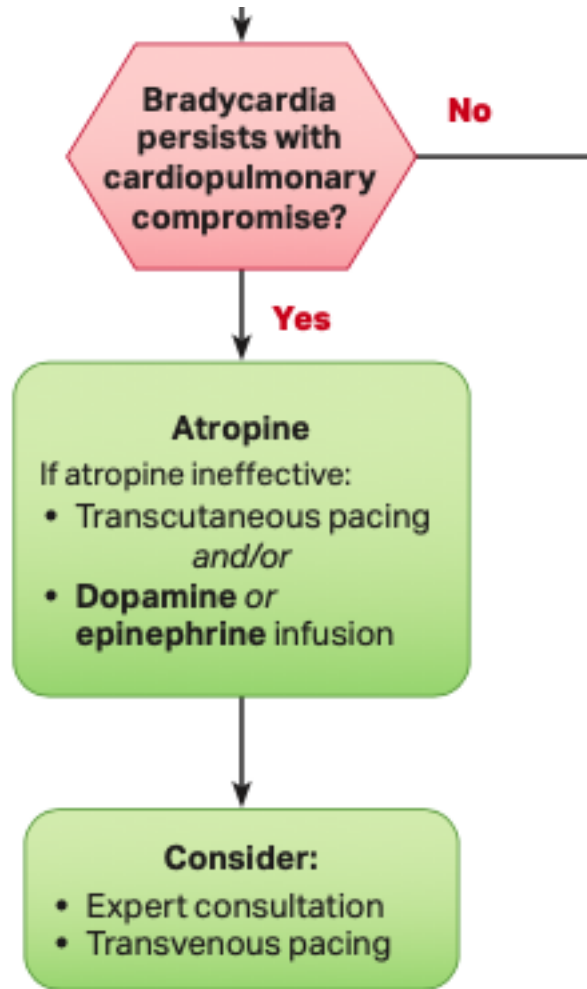
Side effect

- Tachycardia
- Urinary retention
- Toxic doses (atropine > 5 mg) lead to restlessness and excitement, hallucinations, delirium and coma

Administration

- Administer **undiluted by rapid IV injection**
- Slow injection may cause a paradoxical bradycardia

BRADYCARDIA



Atropine IV push
maximum dose 3 mg (5 amp)
ineffective



Transcutaneous pacing

and/or

Dopamine infusion

or

Adrenaline infusion

BRADYCARDIA

Continuous infusion of dopamine or adrenaline **increase heart rate and contractility**

IV infusion		α_1	β_1	β_2
		BP	↑ HR, CO	BP
Dopamine	3-10 mcg/kg/min	+	+++	+
	10-20 mcg/kg/min	+++	+	+
Adrenaline	2-10 mcg/min	+++	+++	+